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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1

Programme: BE.ECE
Course Code: SECB3011
Course Name: SENSORS AND TRANSDUCERS
Batch: 2024-2028

Max. Marks: 50
Time: 2 Hours
SEM: 3
Date: 12.08.2025

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Interpret on the selection criteria's/parameters based on which sensors can be selected for a particular application.	1(2)
2.	Exemplify the different types of sensors.	1(3)
3.	Distinguish a Transducer from that of a Sensor.	1(4)
4.	Examine the use of NQR Thermometry type sensor.	1(4)
5.	Explain the working principle of a humidity sensor.	2(4)
6.	analyze the various applications and advantages of Hot wire anemometer	2(4)
7.	Appraise how (stress) pressure change is measured using a Pizeo-resistive sensor.	2(6)

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Appraise on the concept of triangulation used in GPS and how using it the position and location of an object can be tracked, with necessary illustrations.	1(6)
	(OR)	
9.	Analyze in detail the construction, working principle, advantages, limitations and applications of (a) Acoustic based temperature sensor and (b) Thermo-EMF sensors with neat diagrams.	1(4)
10.	Explain how a sound wave based SONAR can be used for navigation and finding the range of a distance ship in detail with diagrams and necessary expressions.	1(5)
	(OR)	
11.	As an instrumentation engineer, you are tasked with the design of an exhaust gas flow measurement system for a JET engine manufacturing industry. In this regard evaluate on the use of Hot wire anemometers in detail with neat diagrams.	2(6)

12.	Elucidate on the use of potentiometer in generating an equivalent electrical voltage for the physical signal being measured in detail with necessary illustrations. Also list its industrial applications, advantages.	2(5)
	(OR)	
13.	Enumerate how a strain gauge can be used to measure the change in pressure/load being applied with necessary diagrams. Also list its advantages, disadvantages and applications.	2(5)